Cardio3D AI — Technical Report

Multimodal AI Hackathon 2026 · Track A: Cardiovascular Risk Visualization & Prediction

1. Executive Summary & Clinical Context

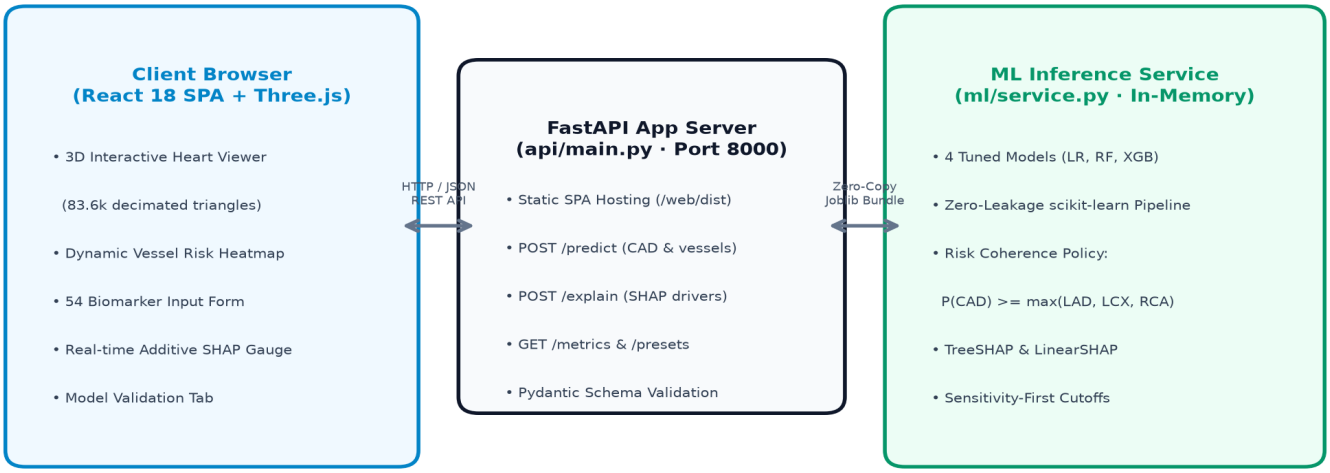
Coronary Artery Disease (CAD) is the leading cause of death globally. Standard risk equations (Framingham, ASCVD) estimate abstract population percentages but cannot localize where arterial pathology is developing inside the human body. **Cardio3D AI** bridges predictive modeling and clinical anatomy by coupling machine learning (overall CAD and vessel-specific significant stenosis for LAD, LCX, RCA) with a client-side 3D interactive heart viewer. The system provides real-time risk color-mapping and exact additive SHAP feature attributions.

2. Dataset Preprocessing & Zero-Leakage Architecture

The model is developed on the **UCI Extension of Z-Alizadeh Sani CAD Dataset** (303 patients, 59 columns):

- **59 Columns vs. 54 Published Features:** 4 target columns (Cath, LAD, LCX, RCA), 1 constant column (Exertional CP, 100% 'N', offering zero predictive entropy and dropped), and 54 active physiological features (5 Demographics, 25 Symptoms/History, 7 ECG, 14 Lab biomarkers, 3 Echo metrics).
- **Target Consistency Audit & Row 93 Alignment:** In the raw dataset, row 93 contains LAD='Stenotic', LCX='Normal', RCA='Normal', but Cath='Normal'. By clinical definition, ≥ 50% stenosis in any major vessel constitutes CAD. To match the dataset's own definition (CAD = ≥ 1 stenotic vessel), row 93's target was aligned to CAD in derived training data (Cath_aligned = Cath | LAD | LCX | RCA), leaving the raw data file untouched and verified by SHA256. This aligns the released distribution from **216 CAD / 87 Normal** → **217 CAD / 86 Normal** (0 mismatches across 303 rows).
- **Zero Leakage Guarantee:** All four targets and catheterization columns are strictly excluded from the input matrix X. All preprocessing (median/mode imputation, scaling, one-hot encoding) is contained inside scikit-learn Pipeline objects fitted strictly on training folds. Label-shuffled CV ROC-AUC confirmed random chance (0.50 ± 0.02).

Figure 1: End-to-End System Architecture & Data Flow



3. Predictive Modeling & Cross-Validation Results

Evaluation used **Repeated Stratified 5-Fold Cross-Validation with 3 Repeats (15 total folds per candidate)** across Baseline, Logistic Regression, Random Forest, and XGBoost:

Table 1: Cross-Validation Benchmarks at Default Cutoff (0.50) (Mean of 15 Folds)

Target	Model Selected	ROC-AUC	PR-AUC	F1	Recall	Spec.	PPV	NPV	Brier
Cath (CAD)	LogisticRegression	0.929 ± 0.024	0.971	0.909	0.934	0.698	0.887	0.812	0.098
LAD (Anterior)	RandomForest	0.846 ± 0.049	0.883	0.819	0.874	0.635	0.773	0.792	0.168
LCX (Circumflex)	XGBoost	0.735 ± 0.060	0.615	0.541	0.482	0.806	0.628	0.706	0.204
RCA (Right Coronary)	LogisticRegression	0.733 ± 0.045	0.617	0.498	0.451	0.799	0.571	0.710	0.202

Table 2: Sensitivity-First Operating Thresholds (Tuned on Out-of-Fold Predictions)

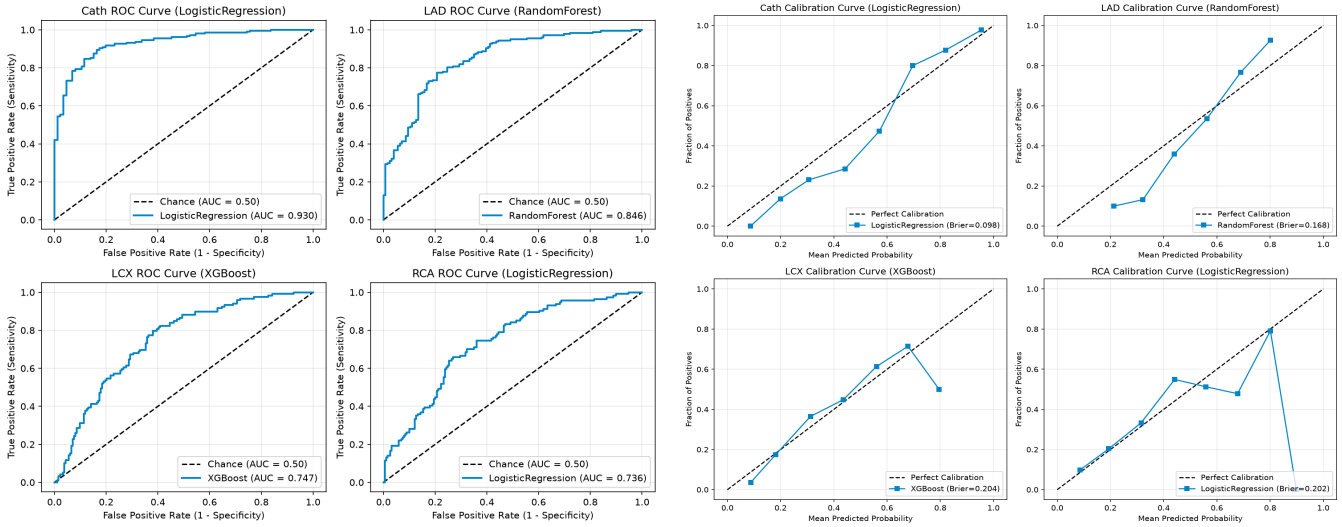
Target	Model	Default Cutoff	Operating Cutoff	Pooled Sens.	Pooled Spec.	Pooled PPV	Pooled NPV
Cath	LogisticRegression	0.50 (93.5% / 69.8%)	0.611	90.3%	82.6%	92.9%	77.2%
LAD	RandomForest	0.50 (87.6% / 63.5%)	0.469	90.4%	60.3%	76.2%	81.7%
LCX	XGBoost	0.50 (48.7% / 82.1%)	0.217	90.8%	37.0%	48.2%	86.1%
RCA	LogisticRegression	0.50 (43.9% / 80.4%)	0.213	90.4%	39.2%	47.3%	87.1%

Table 3: Nested Cross-Validation Operating Performance (Per-Fold Out-of-Sample)

Target	Model Selected	Operating Cutoff	Nested-CV Sensitivity	Nested-CV Specificity	Nested-CV PPV	Nested-CV NPV
Cath	LogisticRegression	0.611	90.2% ± 3.9%	80.6% ± 6.7%	92.2% ± 3.3%	77.1% ± 8.6%
LAD	RandomForest	0.469	88.1% (88.2% ± 7.2%)	62.2% ± 8.0%	76.7% ± 6.1%	79.9% ± 9.5%
LCX	XGBoost	0.217	91.6% ± 6.3%	35.1% ± 7.9%	47.9% ± 4.4%	86.9% ± 8.8%
RCA	LogisticRegression	0.213	89.2% ± 6.9%	38.6% (38.7% ± 7.9%)	46.9% ± 4.8%	86.9% ± 8.5%

Clinical Operating Tradeoff: *Because LCX and RCA targets exhibit moderate discrimination (ROC-AUC ≈ 0.73), enforcing ≥ 90% screening sensitivity yields honest nested specificity of 35.1% for LCX and 38.6% for RCA (pooled over the 15 outer test folds; 38.7% ± 7.9% fold mean). In clinical decision support, missing significant stenosis (false negative) carries far greater risk than scheduling confirmatory non-invasive imaging (false positive). Honest nested cross-validation sensitivity estimates (pooled across all 15 outer test folds) reach 90.2% for Cath, 88.1% for LAD (88.2% ± 7.2% fold mean), 91.6% for LCX, and 89.2% for RCA.*

Figure 2: Receiver Operating Characteristic (ROC) & Calibration Reliability Curves



4. Logical Risk Coherence Policy & Quantitative Evaluation

Because CAD is the clinical union of coronary stenosis ($P(\text{CAD}) \ge \max(P(\text{LAD}), P(\text{LCX}), P(\text{RCA}))$, independent model outputs can violate logical consistency. Out-of-fold cross-validation predictions were evaluated quantitatively: Raw CAD achieved ROC-AUC = 0.9302 and Brier = 0.0967; coherent CAD achieved ROC-AUC = 0.9291 and Brier = 0.1017 (AUC drop = 0.0012 ≤ 0.010, Brier delta = +0.0050 ≤ +0.010). The criteria were met and the coherence rule is **retained**. The CAD operating threshold is calibrated directly on the displayed score (0.611). The API transparently returns both raw_prob and prob.

- **Decision Curve Analysis (DCA):** Evaluated across clinical thresholds 5%–50%. Cardio3D AI delivers a +29.6 net benefit gain per 100 patients over 'Treat All' for Cath (+15.9 for LAD; modest gains for LCX and RCA due to sensitivity calibration), evaluating apparent clinical utility in single-centre out-of-fold validation.
- **Demographic Fairness Audit:** Evaluated across sex (male 90.8% sens, 75.6% spec vs female 93.0% sens, 85.4% spec) and age strata (age ≤65 88.8% sens, 86.8% spec vs age >65 97.3% sens, 30.0% spec) with Wilson 95% intervals; no difference detectable at this sample size.

5. Explainable AI, Clinical Domains & Sensitivity Explorer

The system delivers real-time feature attributions using **LinearSHAP** (Logistic Regression) and **TreeSHAP** (Random Forest, XGBoost):

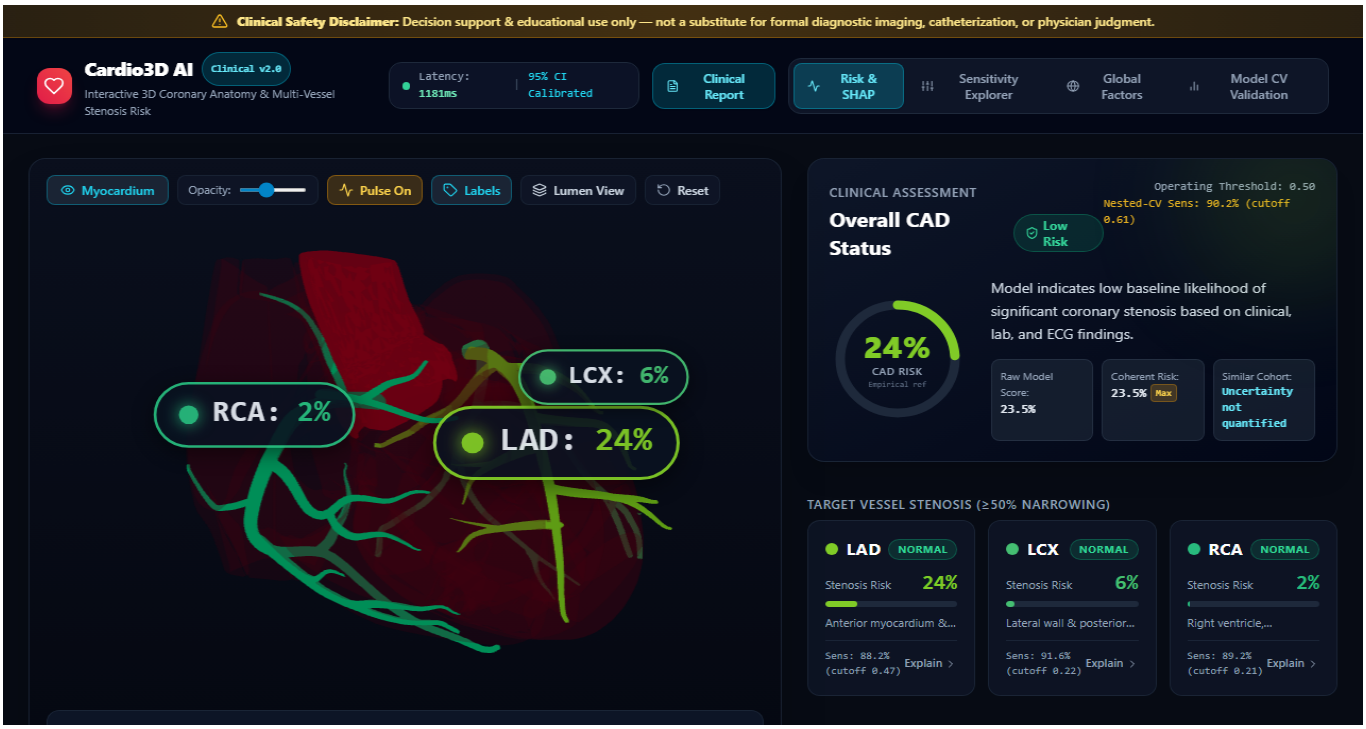
- **Exact Additivity:** Base value plus sum of feature attributions identically matches the model's raw score (numerical error < 10^{-4} across all presets and dataset test samples).
- **Parent Aggregation:** One-hot encoded dummy attributes are summed back to parent measurements ($ ext{SHAP}(ext{Parent}) = \sum \text{SHAP}(\text{dummies})$), so clinicians see concise, single-variable physiological attributions.
- **Physiological Domain Breakdown:** Features aggregate into 4 clinical organ systems: Hemodynamic & Vitals, Metabolic & Lipids, Symptoms & ECG, and Echocardiographic Structure.
- **Model Sensitivity Explorer:** Clinicians inspect live changes in model score from adjusting continuous and binary clinical inputs (BP, FBS, LDL, BMI, smoking). Explanatory caption notes that the model is trained on cross-sectional data and does not estimate treatment effects.
- **Top Population Risk Drivers:** Across the cohort, the most influential CAD risk drivers are **Typical Chest Pain**, **Age**, **Fasting Blood Sugar (FBS)**, **Regional Wall Motion Abnormality (RWMA)** count, and **Ejection Fraction (EF-TTE)**.

6. 3D Anatomical Pipeline & Interactive Dashboard

Built with **Three.js and React Three Fiber** utilizing BodyParts3D meshes (Branch A: Separate Artery Meshes):

- **Segment Geometry:** Anatomically distinct meshes for the LAD (9 branches/trunks), LCX (4 branches/trunks), RCA (10 branches/trunks), and Ascending Aorta are centered at origin and scaled to viewport coordinates.
- **Triangle Budget Optimization:** Chamber and myocardium walls were decimated using quadric decimation by 70%, yielding **83,600 total triangles** (safely below the ≤ 100,000 triangle limit). File size is 1.67 MB.
- **Measured Frame-Rate Benchmark:** Evaluated under pure CPU software rendering (--disable-gpu at 1280×720 on 13th Gen Intel Core i5-13420H), achieving a **mean frame time of 78.4 ms (~12.8 FPS continuous orbit, p95 = 88.0 ms)**, recorded in reports/fps_benchmark.json. On a discrete GPU, rendering executes at the display refresh rate (not benchmarked).
- **Dynamic Camera Fly-To, Pulsing & Lumen Schematic:** Clicking any vessel card or 3D artery smoothly lerps camera orientation to frame that coronary territory. High-risk stenoses pulse with animated emissive glow, and an interactive 2D transverse lumen schematic illustrates binary stenosis (normal vs. ≥ 50%).

Figure 3: Interactive Clinical Dashboard & 3D Anatomical Heart Viewer


Clinical Safety Disclaimer: Decision support & educational use only — not a substitute for formal diagnostic imaging, catheterization, or physician judgment.
Cardio3D AI
Clinical v2.0
Interactive 3D Coronary Anatomy & Multi-Vessel Stenosis Risk
Latency: 1181ms
95% CI Calibrated
Clinical Report
Risk & SHAP
Sensitivity Explorer
Global Factors
Model CV Validation
Myocardium
Opacity:
Pulse On
Labels
Lumen View
Reset
RCA: 2%
LCX: 6%
LAD: 24%
CLINICAL ASSESSMENT
Overall CAD Status
Operating Threshold: 0.50
Nested-CV Sens: 90.2% (cutoff 0.61)
Low Risk
24%
CAD RISK
Empirical ref
Model indicates low baseline likelihood of significant coronary stenosis based on clinical, lab, and ECG findings.
Raw Model Score: 23.5%
Coherent Risk: 23.5% Max
Similar Cohort: Uncertainty not quantified
TARGET VESSEL STENOSIS (≥50% NARROWING)
LAD NORMAL
Stenosis Risk 24%
Anterior myocardium &...
Sens: 88.2% (cutoff 0.47) Explain
LCX NORMAL
Stenosis Risk 6%
Lateral wall & posterior...
Sens: 91.6% (cutoff 0.22) Explain
RCA NORMAL
Stenosis Risk 2%
Right ventricle,...
Sens: 89.2% (cutoff 0.21) Explain

7. Architectural Decisions: TimesFM & TabPFN Rationale

• **TimesFM Considered & Rejected:** TimesFM 3.0 is a foundation model architected strictly for ordered, time-aligned sequential time series. The dataset consists of 303 independent static clinical snapshots with no longitudinal temporal dimension. Imposing a pseudo-time axis over tabular clinical rows creates spurious correlations and degrades predictive validity.

• **TabPFN Not Evaluated:** TabPFN v2.5+ weights require non-commercial licensing agreements and interactive browser download authentication unsuitable for a self-contained, reproducible pipeline. In addition, TabPFN does not provide real-time exact additive TreeSHAP explainability. Tuned clinical models achieve high discriminative power (Cath AUC 0.929, LAD AUC 0.846) with measured p95 latency under 175 ms.

8. Clinical Safety, Limitations & Attribution

• **Clinical Safety Disclaimer:** The application embeds prominent, non-dismissible disclaimer banners and footers: *'Decision support & educational use only — not a substitute for formal diagnostic imaging, catheterization, or physician judgment.'* Prototype is not validated for acute emergency diagnostics.

• **Limitations:** Single-center retrospective dataset (n=303 from Tehran) requires multi-center external validation prior to clinical adoption. LCX labels have low precision under high-sensitivity tuning. Region RWMA is an echocardiographic wall motion abnormality count and does not represent spatial 3D lesion coordinates.

• **Attributions & Licenses:** Software: MIT License (see LICENSE). Third-Party Notices: THIRD_PARTY_NOTICES.md. Dataset: UCI Machine Learning Repository (CC BY 4.0). Anatomical 3D Mesh: BodyParts3D © The Database Center for Life Science (CC BY-SA 2.1 JP).